

NABA HASSAN

SOFTWARE ENGINEER | AI DEVELOPER

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Languages: English and Urdu

PROFESSIONAL EXPERIENCE

AI Engineer | Beyond Eris Solutions

Jan 2026 - Current | Remote

- Architected and deployed enterprise-grade AI production systems using high-performance RAG pipelines, sustaining a 1.18s average response latency on A100 GPU infrastructure.
- Resolved a critical 60% system latency bottleneck by auditing and re-engineering system prompt constraints, execution flow logic, and sampling parameters for the Qwen model series.
- Maintained active R&D; pipelines covering advanced retrieval optimization techniques—including Reciprocal Rank Fusion (RRF) and sparse-dense vector filter parity—while implementing automated multi-stage content processing workflows.

AI/ML Developer Intern | Neusco

July 2025 - Sept 2025 | Karachi, Pakistan

- Developed deep learning networks (CNNs, ANNs, LSTMs) and clean text training data using TensorFlow and Scikit-learn.
- Configured classic classification systems (Random Forest, SVM, KNN) with metric dashboards, and developed cross-platform mobile apps featuring real-time API state syncing using Flutter.

PROJECT EXPERIENCE

Enterprise Multi-Platform Connectors & Routing System (Ownify AI) — Production Project

- Engineered a production-hardened Cascaded Layered Intent Router with a zero-latency Regex Fast-Path, embedding-based Cosine Similarity Router, and structured LLM extraction fallbacks.
- Designed and built real-time, read-only Model Context Protocol (MCP) bridges connecting conversational AI with Google Workspace, Slack, and Notion APIs.
- Built a unified, cross-platform streaming document utility natively supporting text and tabular extraction from PDFs, DOCX, XLSX, PPTX, and CSV files for Google Drive and OneDrive.
- Implemented a "Sticky Context" memory module preserving targeted attachment state variables across multi-turn conversation boundaries.
- Formulated architectural expansion blueprints mapping out read-only connections for Microsoft 365 Graph REST integrations.

Advanced Conversational AI & Contextual RAG System — Core Infrastructure

- Engineered advanced RAG infrastructure powered by LangGraph for multi-agent orchestration and Qdrant for vector space retrieval.
- Applied Reciprocal Rank Fusion (RRF) paired with a BGE-reranker module to eliminate retrieval data anomalies and maintain 95%+ out-of-scope query detection confidence.
- Integrated vLLM and llama.cpp runtimes on Azure cloud, maintaining memory stability and low operational costs via strategic CPU offloading.
- Developed a standalone Natural Language Inference (NLI) validation checkpoint and content cache tracking layer to minimize hallucinations inside sensitive datasets.

Dental Medicine Chatbot & NLP Email Classification System — Independent Projects

- Built an offline conversational dental medicine assistant utilizing Python, SQLite, LangChain, and a locally hosted Meta-Llama-3 LLM.
- Configured intent-detection modules, dynamic SQL generation, and background query caching to provide instant lookup information regarding drug manufacturing.
- Constructed a complete machine learning workflow to categorize communications using classical ML algorithms, deploying via a Streamlit application.

EDUCATION

PGD in Data Science with Artificial Intelligence | NED University of Engineering and Technology

Dec 2025 - Current

Core Focus: Deep Learning, Neural Networks, Knowledge Based Systems, Business Intelligence, Data Visualization, and Digital Forensics.

Bachelors in Software Engineering | Eastern Mediterranean University

Sept 2021 - July 2025 | North Cyprus, Turkey

Honors: Earned prestigious High Honors certifications across consecutive academic terms.

TECHNICAL SKILLS MATRIX

Languages & Core Web Python, C, Urdu, English, HTML5, CSS3, Responsive UI Design.

AI & Orchestration LangGraph, LangChain, Multi-Agent Coordination, Intent Routing, Flow Engineering, Custom Prompt Design, Model Fine-Tuning.

Data Science & ML TensorFlow, Scikit-learn, PyTest, Pandas, NumPy, Keras, PyTorch, Convolutional & Recurrent Neural Networks (CNN/LSTM).

Databases & Vector Qdrant Vector DB, FAISS, Redis Cache, MySQL, SQLite, PostgreSQL.

Tooling & Cloud Model Context Protocol (MCP), Microsoft Graph API, Google Workspace REST API, Slack API, Notion API, OAuth 2.0, Azure GPU Architecture, vLLM Engine, Docker, Git Version Control, n8n Automation, FastAPI.

CERTIFICATIONS

- Machine Learning Specialization - NumPy, Scikit-learn, TensorFlow, and Reinforcement Learning
- IBM AI Developer Professional Certificate - AI-powered web applications, Generative AI, and SDLC
- Deep Learning Specialization - DeepLearning.AI
- Generative AI with Large Language Models - AWS & DeepLearning.AI